

SHIVA KUMAR INDURI

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SUMMARY

Geoinformatics student with an AI/ML minor, skilled in Python, Google Earth Engine, and remote sensing, focused on sustainable infrastructure and precision agriculture.

EDUCATION

Jawaharlal Nehru Technological University Hyderabad (JNTUH) *Aug 2024 – May 2027*
B.Tech in Geoinformatics (Major) | Current CGPA: 8.53/10
AI & Machine Learning (Minor) | Current CGPA: 8.0/10

S.G. Govt Polytechnic, Adilabad *June 2021 – April 2024*
Diploma in Civil Engineering | Overall CGPA: 8.77/10
Awarded the **Meritorious Student Award** for academic excellence in Civil Engineering.

TECHNICAL EXPERTISE

- **Geospatial Analysis:** QGIS, ArcGIS Pro, Google Earth Engine (GEE), Bhuvan Geoportal, PostGIS.
- **Programming:** Python (NumPy, Pandas, GeoPandas, Rasterio, OpenCV), SQL (PostgreSQL/MySQL), JavaScript (GEE API).
- **Development:** html,css,Git/GitHub.

PROJECTS

- Hydro Harvest AI: Automated Geospatial Design Suite** *March 2026 - Present*
- **Innovation:** Engineered a software-only suite to automate Rooftop Rainwater Harvesting (RWH) designs using high-resolution satellite imagery.
 - **Methodology:** Developed a GIS-driven pipeline using React Native, Django, and PostGIS for automated rooftop boundary extraction and runoff modeling.
 - **Impact:** Integrated 30-year historical rainfall datasets and soil permeability layers to optimize storage tank sizing—a software-only approach that won the Jal Shakti Hackathon 2025 (Ministry of Jal Shakti, Govt. of India).

- Urban Heat Island (UHI) Spatio-Temporal Analysis** [live](#) *Hyderabad, 2024*
- **Analysis:** Processed **Landsat 8/9** TIRS data in GEE to quantify surface urban heat island (SUHI) intensity across the metropolitan region.
 - **Algorithm:** Developed a spatial buffer algorithm to analyze the correlation between urban density (NDBI), vegetation (NDVI), and thermal retention (LST).
 - **Result:** Confirmed a strong negative correlation between NDVI and LST, providing a data-driven model for urban cooling interventions.

EXPERIENCE

- India Space Lab | Technical Trainee (Winter Internship)** *Feb 2026 – March 2026*
- **Satellite Systems:** Prototyped sensor-to-satellite data pipelines for **CanSat** and **CubeSat** architectures.
 - **Earth Observation:** Leveraged Remote Sensing and GIS for agricultural optimization and disaster management modeling.
- Govt. of Telangana, Irrigation & CAD Dept | Industrial Trainee** *Dec 2023 – June 2024*
- **Hydrology:** Conducted field-based hydraulic surveys and monitored water distribution for state-level irrigation projects.
 - **Data Management:** Streamlined technical documentation for irrigation command area development and infrastructure monitoring.

VOLUNTEER EXPERIENCE & TRAINING

- **UN Mappers Mapathon:** Mapped forest fire-prone zones in Adilabad on March 18, 2026.
- **NRSC Training:** Specialized training on the **Bhuvan Geoportal** and SAR data (NISAR) trends at ISRO Shadnagar.